

2 Derivatives and Integrals with Power Series

Goal: Find a power series for $\frac{1}{(1-x)^2}$

First, observe that

$$\frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{d}{dx} \left((1-x)^{-1} \right) = -(1-x)^{-2}(-1) = \frac{1}{(1-x)^2}$$

Plan: Get the power series for $\frac{1}{1-x}$, then take its derivative. $a=1$ $r=x$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

$\frac{d}{dx}$ $\swarrow$

$$\frac{1}{(1-x)^2} = 0 + 1 + 2x + 3x^2 + 4x^3 + \dots = \boxed{\sum_{n=0}^{\infty} nx^{n-1}}$$

2.0.1 Example:

Find the power series representation for $\arctan(x)$, and find its interval of convergence.

$$\left(\text{Hint: } \frac{d}{dx} \left(\arctan(x) \right) = \frac{1}{1+x^2} \right)$$

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = 1 - x^2 + x^4 - x^6 + \dots = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

(Note: A pink arrow labeled "integrate" points from the first fraction to the second. Orange arrows point from the x^2 and x^4 terms in the series to the corresponding terms in the integrated series below. A pink note $(-1)^n \cdot (x^2)^n$ is written below the summation.)

$$\arctan(x) = K + x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots = K + \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$\arctan(0) = K + 0 - \frac{1}{3}0^3 + \frac{1}{5}0^5 - \dots$$

$$0 = K$$

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

- $f^{(4)}(x) = 2 \cdot 3 \cdot 4 c_4 x + \dots = e^x$

In general, the coefficient c_n in the Maclaurin Series of $f(x)$ is

$$c_n = \frac{f^{(n)}(0)}{n!}$$

Maclaurin Series

Function	Series	ROC
$\frac{1}{1-x}$	$= \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$	$R = 1$
e^x	$= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	$R = \infty$
$\sin x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	$R = \infty$
$\cos x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	$R = \infty$
$\arctan x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$	$R = 1$
$\ln(1+x)$	$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$	$R = 1$

$$\begin{aligned}
 & \frac{d}{dx} \left(e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \right) \\
 & \downarrow \\
 & e^x = 0 + 1 + x + \frac{1}{2}x^2 + \dots
 \end{aligned}$$

2 Examples

In the following examples, use the table of Maclaurin series to write out at least 4 nonzero terms of the Maclaurin series for each function.

2.0.1 Example:

$$f(x) = x \sin x \quad \textcolor{violet}{x} \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \cdot \textcolor{violet}{x}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+1)!}$$

2.0.2 Example:

$$f(x) = \sin(x^3) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^3)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+3}}{(2n+1)!}$$

2.0.3 Example:

$$f(x) = e^{3x}$$

2.0.4 Example:

The integral $\int e^{x^2} dx$ cannot be integrated using elementary functions. Find the Maclaurin polynomial for this integral with at least four nonzero terms.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\int e^{x^2} dx = \int \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} dx = \int \sum_{n=0}^{\infty} \frac{x^{2n}}{n!} dx$$

$$\int e^{x^2} dx = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1) \cdot n!}$$

3 Taylor Series

Taylor series are like Maclaurin Series, but can be centered anywhere—not just at 0. The Taylor Series for $f(x)$, centered at a is

$$f(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \cdots \quad \text{for } |x - a| < R,$$

where R is the radius of convergence.

The coefficient c_n in the Taylor Series of $f(x)$ is

$$c_n = \frac{f^{(n)}(a)}{n!}$$